Urticaria

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Definition

Urticaria is a condition characterized by the development of wheals, angioedema, or both.

A wheal has three typical features:

1. Sharply circumscribed superficial central swelling of variable size and shape, almost invariably surrounded by reflex erythema
2. Itching or sometimes burning sensation
3. Resolves usually within 30 minutes to 24 hours.

Angioedema is characterized by:

- ☑ Sudden, erythematous, or skin-colored deep swelling in the lower dermis and subcutaneous plane or mucous membranes
- ☑ Tingling, burning, tightness, and sometimes pain rather than itch
- ☑ Resolution can take up to 72 hours (slower than that of wheals).

Differences between acute urticaria and chronic urticaria are presented in **Table 1**.

TABLE 1: Differences between acute urticaria and chronic urticaria.

Acute urticaria	***Chronic urticaria***
☑ Lasts for <6 weeks ☑ Usually due to allergy to food, medications, insect stings, infections, or transfusion reactions	☑ Lasts for >6 weeks ☑ Includes physical urticaria, idiopathic urticaria and urticaria in autoimmune diseases, neoplasms, and autoinflammmatory diseases

Causes

Acute urticaria may occur by immunoglobulin E (IgE)-mediated or non-IgE-mediated pathway ***(Fig. 1)***.

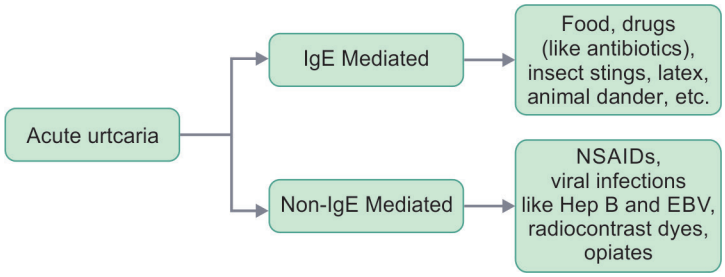

Fig. 1: Causes of acute urticaria.
(EBV: Epstein–Barr virus; NSAIDs: nonsteroidal anti-inflammatory drugs)

Common infections producing acute urticaria are shown in ***Figure 2***.

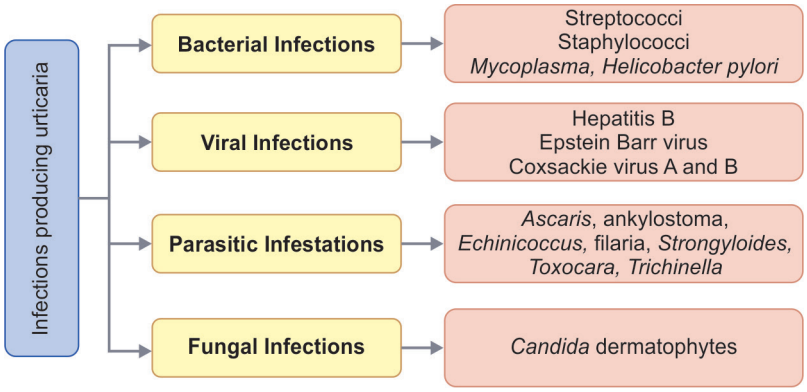

Fig. 2: Common infections producing acute urticaria.

Urticarial Vasculitis

Autoinflammatory diseases like systemic lupus erythematosus (SLE) can produce urticarial vasculitis. These lesions have a burning sensation (rather than itching), will not blanch, last beyond 24 hours, and heal by scarring. Urticarial vasculitis should be differentiated from urticaria.

Chronic Urticaria (Fig. 3)

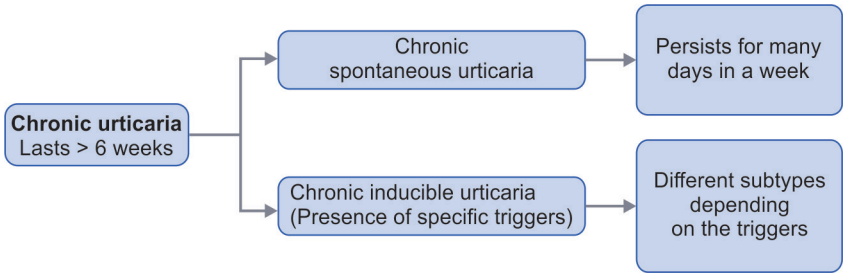

Fig. 3: Chronic urticaria.

Chronic Spontaneous/ Idiopathic Urticaria

Chronic urticaria, which is not related to allergic reaction, may be associated with the presence of antithyroid antibodies, especially antithyroid peroxidase (TPO) antibodies. 35–40% may have a positive autologous serum skin test (pathergy test). Some patients may have anti-IgE or anti-IgE receptor antibodies.

Chronic Inducible Urticaria

The different types of physical urticaria are given in the below boxes.

Cold urticaria	Cholinergic urticaria	Pressure-induced urticaria
Intense itching, erythema, and urticaria on exposure to cold. Isomorphic cold reaction can be elicited by keeping ice cube on skin for 5 minutes.	Small punctate pruritic urticaria, surrounded by flare. Occurs after exercise, hot showers, or sweating. Resolves in 30–60 minutes.	Symptoms appear 4–6 hours after pressure was applied or on dependent body parts. Can coexist with other physical urticara.

Solar urticaria	Aquagenic urticaria	Dermatographism
Occurs within minutes of sun exposure. Subsides within 1–3 hours after stopping the exposure. Six different types based on the wavelength.	Urticaria occurring on exposure to water, irrespective of the temperature. Direct exposure of skin to water is used for testing for this.	Also known as urticaria facticia. Ability to write on skin using blunt objects. Can be isolated disorder or associated with physical urticaria.

Diagnosis

Diagnosis is mainly clinical. Detailed history will help in identifying the inciting event or cause and will help in making a correct diagnosis.

Keeping a food/event diary may help in identifying the allergen in patients with recurrent urticaria.

Allergy skin testing is not usually indicated for patients with acute urticaria.

If chronic idiopathic urticaria is suspected, test for anti-TPO antibodies and do autologous serum skin test.

Testing for different types of physical urticaria has to be done if the history is suggestive of physical/inducible urticaria.

Treatment of Acute Urticaria

Antihistamines

- ☑ Antihistamines are the first-line drugs used.
- ☑ Second-generation antihistamines, which are less sedating, are preferred whenever sedation is a concern such as older school going children and adolescents.
- ☑ However, in infants and toddlers, first-generation may be used to reduce irritability and improve sleep.
- ☑ The dosages of antihistamines are given in **Table 2**.

TABLE 2: Dosage of antihistamines.

Drug	***Dosage***
Hydroxyzine	2 mg/kg/day in three divided doses
Diphenhydramine	0.5 mg/kg/dose Q6–8 hourly
Cetirizine	2–5 years—2.5 mg OD, 6–11 years—5 mg OD, >12 years—10 mg OD
Levocetirizine	2–5 years—1.25 mg OD, 6–11 years—2.5 mg OD, >12 years—5 mg OD
Fexofenadine	6–11 years—30 mg BD, >12 years—60 mg BD
Loratadine	2–5 years—5 mg OD, >6 years—10 mg OD

Adrenaline

It is used for extensive urticaria and angioedema. Dose—injection adrenaline 0.01 mL/kg of 1/1,000 adrenaline solution, subcutaneous or intramuscular.

Oral Steroids

It is indicated only in severe urticaria or urticaria not responding to antihistamines. Tablet prednisolone 1–2 mg/kg/day for 3–5 days.

Treatment of Chronic Urticaria

Second-generation antihistamines are the main stay of management **(Fig. 4)**.

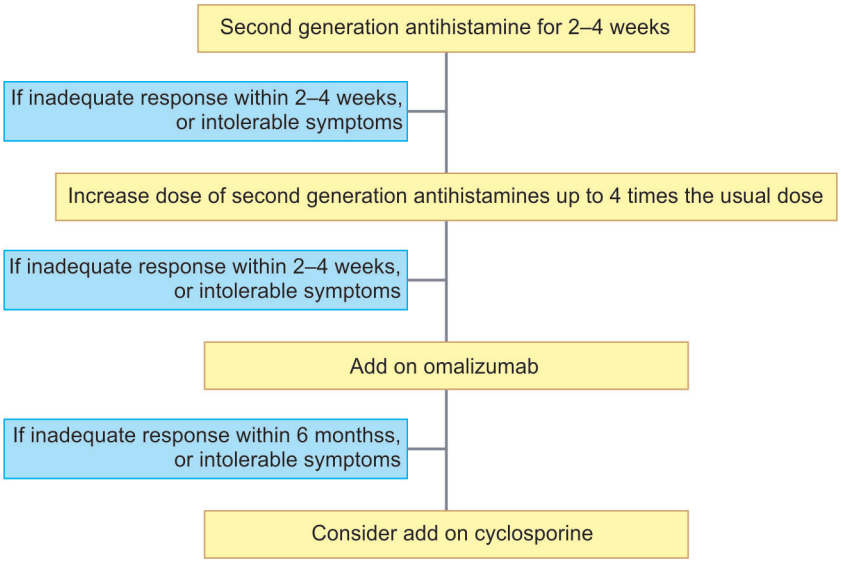

Fig. 4: Management of chronic urticaria.

A short course (3–5 days) of oral steroids may be used in a severely symptomatic patient. But steroids are not recommended for long-term treatment of chronic urticaria.

Further Reading

- ☑ Godse K, De A, Zawar V, Shah B, Girdhar M, Rajagopalan M, et al. Consensus statement for the diagnosis and treatment of urticaria: a 2017 update. Indian J Dermatol. 2018;63(1):2-15.
- ☑ Zuberbier T, Aberer W, Asero R, Abdul Latiff AH, Baker D, Ballmer-Weber B, et al. The EAACI/GA^{2}LEN/EDF/WAO guideline for the definition, classification, diagnosis and management of urticaria. Allergy. 2018;73(7):1393-414.
- ☑ Zuberbier T, Abdul Latiff AH, Abuzakouk M, Aquilina A, Asero R, Baker D, et al. The international EAACI/GA^{2}LEN/EuroGuiDerm/APAAACI guideline for the definition, classification, diagnosis, and management of urticaria. Allergy. 2021;00:1-33.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Acute Otitis Media

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